

Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization

A PROJECT REPORT

Submitted by,

Mr. G Hemanth Kumar	-20211CSD0073
Mr. Guddattu Ravith Kumar	-20211CSD0128
Mr. Abhishek Shivilal Rathod	-20211CSD0046
Mr. Salmondavid F Potagoli	- 20221LSD0007

Under the guidance of,

Ms. Ankita Bhaumik

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING, COMPUTER ENGINEERING,
INFORMATION SCIENCE AND ENGINEERING Etc.**

At



PRESIDENCY UNIVERSITY

BENGALURU DECEMBER 2024

PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

CERTIFICATE

This is to certify that the Project report “Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization”
being submitted by “G Hemanth Kumar, Guddattu Ravith Kumar, Abhishek Shivilal Rathod, Salmondavid F Potagoli ” bearing roll number(s) “20211CSD0073, 20211CSD0128, 20211CSD0046, 20221LSD0007” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a bonafide work carried out under my supervision.

Ms.Ankita Bhaumik
Assistant Professor
School of CSE&IS
Presidency University

Dr.Saira Bhanu
Professor & HoD
School of
CSE&IS
Presidency
University

Dr. L. SHAKKEERA
Associate Dean
School of CSE
Presidency University

Dr. MYDHILI NAIR
Associate Dean
School of CSE
Presidency University

Dr. SAMEERUDDIN KHAN
Pro-Vc School of Engineering
Dean -School of CSE&IS
Presidency University

PRESIDENCY UNIVERSITY

SCHOOL OF COMPUTER SCIENCE ENGINEERING

DECLARATION

We hereby declare that the work, which is being presented in the project report entitled **Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization** in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of our own investigations carried under the guidance of **Ms.Ankita Bhaumik, Assistant Professor, School of Computer Science Engineering & Information Science, Presidency University, Bengaluru.**

We have not submitted the matter presented in this report anywhere for the award of any other Degree.

**Name(s), Roll No(s) and Signature(s) of
the Students**

ABSTRACT

The recruitment process has traditionally been time-consuming, error-prone, and heavily reliant on manual effort. "Smart Hiring: Job Portal with AI-Driven ATS Resume Optimization" introduces an innovative approach to streamline and enhance the hiring experience. This platform integrates an AI-powered Applicant Tracking System (ATS) that automatically optimizes job applicants' resumes for better alignment with job descriptions, improving the accuracy of candidate selection. Leveraging machine learning algorithms, the system analyzes resumes, identifies key skills and keywords, and suggests enhancements to increase visibility and relevance. The job portal, built using modern web technologies, provides an intuitive interface for both recruiters and job seekers. For recruiters, it ensures a more efficient candidate screening process by ranking resumes based on fit, while job seekers benefit from improved resume content and better chances of matching with suitable job openings. By reducing bias, increasing efficiency, and enhancing the candidate experience, this solution fosters smarter, more data-driven hiring decisions

ACKNOWLEDGEMENT

First of all, we indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this project on time.

We express our sincere thanks to our respected dean **Dr. Md. Sameeruddin Khan**, Pro-VC, School of Engineering and Dean, School of Computer Science Engineering & Information Science, Presidency University for getting us permission to undergo the project.

We express our heartfelt gratitude to our beloved Associate Deans **Dr. Shakkeera L and Dr. Mydhili Nair**, School of Computer Science Engineering & Information Science, Presidency University, and Dr. “Saira Bhanu”, Head of the Department, School of Computer Science Engineering & Information Science, Presidency University, for rendering timely help in completing this project successfully.

We are greatly indebted to our guide **Ms.Ankita Bhaumik, Assistant Professor** and Reviewer **Dr./Mr.Ms. Name, Designation**, School of Computer Science Engineering & Information Science, Presidency University for her inspirational guidance, and valuable suggestions and for providing us a chance to express our technical capabilities in every respect for the completion of the project work.

We would like to convey our gratitude and heartfelt thanks to the PIP2001 Capstone Project Coordinators **Dr. Sampath A K, Dr. Abdul Khadar A and Mr. Md Zia Ur Rahman**, department Project Coordinators **“NAME”** and Git hub coordinator **Mr. Muthuraj**.

We thank our family and friends for the strong support and inspiration they have provided us in bringing out this project.

G Hemanth Kumar
Guddattu Ravith Kumar
Abhishek Shivilal Rathod
Salmondavid F Potagoli

LIST OF FIGURES

Sl. No.	Figure Name	Caption	Page No.
1	Figure 6.1	Usecase Diagram	
2	Figure 6.2	DFD 0 level Diagram	
3	Figure 6.3	DFD 1 level Diagram	
4	Figure 6.4	DFD 2 level Diagram	
5	Figure 6.5	Class Diagram	
6	Figure 6.6	ER Diagram	
7	Figure 6.7	Activity Diagram (Employer)	
8	Figure 6.8	Activity Diagram (Employee)	
9	Figure 6.9	Activity Diagram (Admin)	
10	Figure 10	Register Page	
11	Figure 11	Login Page	
12	Figure 12	Admin Dashboard Page	
13	Figure 13	All Users Page	
14	Figure 14	Register a job Page	
15	Figure 15	Jobs List Page	
16	Figure 16	Create Job Category Page	
17	Figure 17	Job Categories Page	
18	Figure 18	Employee Dashboard	
19	Figure 19	Employee Jobs History Page	
20	Figure 20	Employee Personal Info Page	
21	Figure 21	Apply for Job Page	
22	Figure 22	Ats Evaluator	
23	Figure 23	Evaluated Resume	

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	
1.	INTRODUCTION	
	1.1 Brief Overview of Work	
	1.2 The Evolution of Recruitment and Its Challenges	
	1.3 Emergence of AI-Driven Recruitment Solutions	
	1.4 MERN Technology Stack in Job Portals	
	1.5 Impact and Significance	
2.	LITERATURE REVIEW	
	2.1 Importance of Job Portals	
	2.2 Project Feasibility Study	
	2.2.1 Technical Feasibility	
	2.2.2 Economical Feasibility	
	2.2.3 Operational Feasibility	
3.	RESEARCH GAPS OF EXISTING METHODS	
	3.1 Limited Resume Analysis Capabilities	
	3.2 Inability to Address Bias	
	3.3 Scalability and Efficiency Challenges	
	3.4 Poor Candidate Experience	
	3.5 Inadequate Job Matching Algorithms	
	3.6 Limited Use of Predictive Analytics	
4.	PROPOSED METHODOLOGY	
4.1	Requirement Analysis and Design	
4.2	Front-end Development Using React.js	
4.3	Back-end Development with Node.js and Express.js	
4.4	Database Management with	

	MongoDB
4.5	Integration of AI-Driven ATS
4.6	Job Recommendation System
4.7	Testing and Validation
4.8	Continuous Improvement
5.	OBJECTIVES
6.	SYSTEM DESIGN & IMPLEMENTATION
6.1	Usecase Diagram
6.2	DFD 0 level Diagram
6.3	DFD 1 level Diagram
6.4	DFD 2 level Diagram
6.5	Class Diagram
6.6	ER Diagram
6.7	Activity Diagram (Employer)
6.8	Activity Diagram (Employee)
6.9	Activity Diagram (Admin)
8	OUTCOMES
8.1	Enhanced Resume Evaluation
8.2	Accessibility and User-Friendly Interface
8.3	Real-World Job Application Optimization
8.4	Time-Saving for Job Seekers
8.5	Improved Career Awareness and Strategy
9.	RESULTS AND DISCUSSIONS
9.1	Improved Resume Alignment with Job Descriptions
9.2	Enhanced Accessibility and Usability
9.3	Key Insights into Resume Gaps
9.4	Real-World Applicability and Use Cases
9.5	Empowerment Through Career Awareness
10.	CONCLUSION
11.	REFERENCES
12.	PSUEDOCODE
13.	SCREENSHOTS

CHAPTER-1

INTRODUCTION

1.1 Brief Overview of Work

"Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization" is an innovative recruitment platform that combines the scalability of the MERN stack (MongoDB, Express.js, React, Node.js) with the intelligence of AI-powered Applicant Tracking Systems (ATS) to streamline the hiring process. The portal enables recruiters to automate resume screening, rank candidates based on job relevance, and manage applications efficiently through a dynamic dashboard. For job seekers, the AI optimizes resumes by analyzing keywords, skills, and formatting to enhance alignment with job descriptions, improving their chances of success. By integrating modern web technologies with machine learning, this platform addresses inefficiencies in traditional hiring, reduces bias, and creates a seamless experience for all users, making recruitment smarter, faster, and more effective.

1.2 The Evolution of Recruitment and Its Challenges

The recruitment landscape has undergone significant changes over the years, driven by the growth of digital platforms and the increasing demand for skilled talent. However, these advancements have also introduced challenges, such as:

- **Volume Overload:** Recruiters are overwhelmed by the sheer number of applications, making it difficult to identify top candidates quickly.
- **Misaligned Applications:** Job seekers often submit generic resumes that do not match specific job requirements, reducing their chances of selection.
- **Bias and Manual Errors:** Traditional hiring methods can inadvertently introduce biases and are prone to errors due to manual intervention.

1.3 Emergence of AI-Driven Recruitment Solutions

The emergence of AI-driven recruitment solutions marks a transformative shift in the hiring process, leveraging artificial intelligence to enhance efficiency, accuracy, and fairness in recruitment. Traditional methods, such as manual resume screening and

subjective decision-making, often led to time-consuming processes, human bias, and inconsistent hiring decisions. AI-driven solutions address these challenges by automating key aspects of recruitment, from candidate sourcing and resume screening to interview scheduling and final selection. Machine learning algorithms can analyze vast amounts of data, including resumes, job descriptions, and candidate profiles, to identify patterns and predict the best job matches based on skills, experience, and even cultural fit. AI can also optimize resumes for Applicant Tracking Systems (ATS), ensuring that candidates' applications are tailored to meet specific job requirements, increasing their chances of being noticed. Additionally, AI helps reduce unconscious bias by focusing on data-driven insights rather than subjective factors, promoting more inclusive and equitable hiring practices. These advancements not only streamline the recruitment process but also provide recruiters with actionable insights, enhance the candidate experience, and create a faster, more transparent, and efficient hiring ecosystem.

1.4 MERN Technology Stack in Job Portals

The MERN stack (MongoDB, Express.js, React, and Node.js) has emerged as a preferred choice for building scalable and dynamic web applications. In the context of a job portal, it offers:

- **Seamless Frontend and Backend Integration:** React for an intuitive user interface and Node.js with Express.js for efficient server-side operations.
- **Robust Data Management:** MongoDB for flexible and scalable storage of candidate and job data.
- **Real-Time Updates:** Enabling instant feedback and live application status tracking for users.

1.5 Impact and Significance

The impact and significance of AI-driven recruitment solutions lie in their ability to revolutionize traditional hiring practices by enhancing efficiency, fairness, and scalability. These systems automate labor-intensive tasks such as resume screening, candidate ranking, and job matching, significantly reducing time-to-hire and operational costs for recruiters. By leveraging data-driven algorithms, they minimize human bias and

promote equitable hiring practices, ensuring that candidates are evaluated based on skills and qualifications rather than subjective factors. For job seekers, AI-powered platforms provide valuable insights, such as resume optimization and personalized job recommendations, increasing their chances of securing relevant opportunities. Additionally, these solutions improve the overall candidate experience through real-time feedback and transparent application tracking. From a business perspective, organizations benefit from a more streamlined recruitment pipeline, better talent acquisition, and improved employer branding. Ultimately, AI-driven recruitment fosters smarter, faster, and more inclusive hiring processes, addressing key challenges in today's dynamic job market while setting a new benchmark for the future of work

CHAPTER-2

LITERATURE SURVEY

Job Procurement: Old and New Ways Job seeking usually involves different ways to look for jobs such as through personal contacts, direct telephone calls to employers, job agency office, scanning online job listings, etc. Before the Internet, became widely uses as a method of seeking jobs, jobseekers spent a lots of time using various methods to look for job openings. Today, jobseekers use online methods which are very convenient and save a lot of time. Galanaki lists the following methods to be the traditional (old) ways for recruitment:

- 1)Employment recruitment agencies
- 2)Job fairs
- 3)Advertising in the mass media such as newspapers
- 4)Management Consultants
- 5)Advertisement in television and radio
- 6)Existing employee contacts
- 7)Schools colleges or universities students services department
- 8)Workers or professional referrals

These old job seeking methods are too slow, stressful, challenging and also lack quality. In addition, the applicants have to consider the cost and the amount of time to get the information they need, and other preparations they have to make. Finding all available job vacancies is a main step at in the job-seeking process. The Internet is now a powerful tool that jobseekers can use. Today, there are many sites that advertise job positions to be filled by people with certain skills in various fields. The Internet plays an important role in the area of human resource planning and development. Most planning and development organizations are now using computer technology and the Internet for staff recruitment. It should be noted that although the Internet has facilitated the process of job-seeking, it has not replaced the traditional methods, completely.

Importance of Job Portals

In the age of technology, the Internet has become the main source of information for jobseekers. Large corporations, Institutions, and universities include information on career Prospects on their websites. According to a survey, 70% of the workforce uses websites or portals on the Internet to

Search for jobs in France. These websites or portals provide search engine to access information on job opportunities.

2.2 Project Feasibility Study

2.2.1 Technical Feasibility

Technical feasibility study is concerned with specifying equipment and software that will successfully satisfy the user requirement; the technical needs of the system may vary considerably.

The facility to produce outputs in a given time. Our project is a web based application which is based on client-server based application. In this application every page as output is render from server to client so it is necessary that the page should be rendered in time. For this I have avoided more and more code in the page- load event.

2.2.2 Economical Feasibility

Economical feasibility is the measure to determine the cost and benefit of the proposed system. A project is economical feasible which is under the estimated cost for its development. These benefits and costs may be tangible or intangible. Job Portal is the cost-effective project in which there is less possibility of intangible cost so there is no difficulty to determine the cost of the project.

2.2.3 Operational Feasibility

Operation feasibility is used to check whether the project is operationally feasible or not. Our project is mainly different from the other system because of its web-support feature. So the measure for operational feasibility is something different from other system. Generally the operational feasibility is related to organization aspects.

The change determination is as such that early product were either a man or group of men or the jobs based manual but now a day with the advent of Internet technology.

CHAPTER-3

RESEARCH GAPS OF EXISTING METHODS

3.1 Limited Resume Analysis Capabilities

Traditional Applicant Tracking Systems (ATS) predominantly rely on keyword matching algorithms to screen and rank candidates. These systems often fail to account for:

- **Contextual Relevance:** They lack the ability to understand the broader context of a candidate's experience or transferable skills. For example, a candidate with unconventional job titles may be overlooked despite having relevant expertise.
- **Formatting Sensitivities:** Poorly formatted resumes are frequently rejected, even if they contain relevant information.
- **Lack of Feedback Mechanisms:** Job seekers do not receive constructive feedback on why their resumes were rejected or how they can improve alignment with job descriptions.

3.2 Inability to Address Bias

While many systems claim to promote fair hiring practices, they often fail to mitigate unconscious biases:

- **Algorithmic Bias:** The datasets used to train these systems may inadvertently reinforce societal biases, leading to unfair evaluations based on factors like gender, ethnicity, or educational background.
- **Human Oversight:** In hybrid systems where manual screening is involved, personal biases can influence hiring decisions, undermining objectivity.

3.3 Scalability and Efficiency Challenges

Although current systems are designed to handle high volumes of applications, they struggle in certain scenarios:

- **Specialized Roles:** For niche or highly specialized roles, these systems often lack the sophistication to effectively evaluate candidate suitability.
- **High Application Volumes:** Systems may rank candidates inadequately due to their inability to assess nuanced qualifications or compare applicants beyond keywords.

3.4 Poor Candidate Experience

One of the most significant drawbacks of existing methods is the lack of a candidate-centric approach:

- **Limited Transparency:** Applicants are often left uninformed about their application status, which creates frustration and disengagement.
- **Generic Processes:** Most systems provide a uniform experience, failing to tailor the process to individual needs, such as entry-level applicants versus experienced professionals.

3.5 Inadequate Job Matching Algorithms

Job matching algorithms in traditional recruitment systems often fail to provide accurate and meaningful connections between candidates and job opportunities. These algorithms typically rely on rigid keyword-based matching, focusing on explicit terms present in resumes and job descriptions without accounting for the context or nuances of skills and experiences. For instance, a candidate with a wealth of relevant experience but different terminology for job roles or skills might be overlooked. Similarly, the algorithms often fail to account for transferable skills that may be applicable across industries or roles. This lack of contextual understanding results in mismatches, where either unqualified candidates are shortlisted or highly qualified candidates are excluded from the selection pool.

3.6 Limited Use of Predictive Analytics

Existing recruitment tools rarely leverage predictive analytics to forecast:

- **Candidate Performance:** The potential success of a candidate in the role based on historical and behavioral data.
 - **Turnover Risks:** Whether a candidate is likely to stay with the organization long-term.
- This lack of foresight results in higher attrition rates and mismatches in hiring.

Addressing these research gaps requires the integration of advanced AI technologies, personalized candidate experiences, and dynamic algorithms capable of adapting to evolving market needs. A more holistic approach that focuses on fairness, transparency, and efficiency will pave the way for smarter, more inclusive recruitment solutions.

CHAPTER-4

PROPOSED METHODOLOGY

4.1 Requirement Analysis and Design

The foundation of the project begins with identifying the key requirements for both employers and job seekers. Employers need a streamlined method to post job openings, review resumes efficiently, and shortlist candidates quickly. Job seekers require a platform to create profiles, upload resumes, and apply to relevant job openings seamlessly. The project involves designing a scalable architecture that incorporates the MERN (MongoDB, Express.js, React.js, Node.js) stack for robust frontend and backend integration. Special attention is given to the design of an Applicant Tracking System (ATS) with AI capabilities to evaluate and rank resumes based on job-specific criteria.

4.2 Front-end Development Using React.js

The user interface of the job portal is developed using React.js, providing a dynamic and interactive experience. Features such as job search, profile creation, and application tracking are designed for user convenience. The frontend incorporates modern UI/UX principles to ensure accessibility and ease of navigation. Responsive design ensures compatibility across devices, catering to both desktop and mobile users. Components are modular, enabling scalability and reusability.

4.3 Back-end Development with Node.js and Express.js

The backend is implemented using Node.js and Express.js to handle business logic, API integrations, and user authentication. RESTful APIs facilitate communication between the frontend and backend, enabling features such as job posting, resume submission, and application management. Secure user authentication and authorization mechanisms are implemented using JWT (JSON Web Tokens) to safeguard user data and prevent unauthorized access.

4.4 Database Management with MongoDB

MongoDB serves as the database for storing and managing user profiles, job listings, and application details. The database is structured to support efficient querying, enabling quick retrieval of relevant information such as matching job postings for candidates or potential candidates for recruiters. Indexing and schema design focus on optimizing search performance, especially for AI-driven resume evaluations.

4.5 Integration of AI-Driven ATS

The AI-driven Applicant Tracking System is integrated into the job portal to enhance the hiring process. It utilizes Natural Language Processing (NLP) and machine learning algorithms to analyze resumes, extract key details, and match them with job descriptions. The system scores and ranks candidates based on criteria such as skills, experience, and keywords in the job description. The ATS also provides feedback to job seekers on improving their resumes to increase their chances of selection.

4.6 Job Recommendation System

A recommendation engine is developed using AI techniques to suggest suitable jobs to candidates based on their profiles, preferences, and application history. Similarly, employers receive recommendations for candidates who closely match their job requirements. This personalized matching enhances the efficiency of the platform and ensures better outcomes for both parties.

4.7 Testing and Validation

Extensive testing is conducted to ensure the functionality, reliability, and security of the platform. Unit testing, integration testing, and user acceptance testing are performed for each module. The AI-driven ATS is tested with a variety of resumes and job descriptions to validate its accuracy and effectiveness. Feedback from potential users is incorporated to refine the platform before deployment.

4.8 Continuous Improvement

Post-deployment, the platform undergoes iterative improvements based on user feedback and emerging industry trends. The AI models are retrained periodically with new data to improve their accuracy and adaptability. New features, such as advanced analytics for recruiters and personalized career guidance for job seekers, can be added in subsequent versions.

This structured methodology ensures that the job portal is robust, user-friendly, and capable of revolutionizing the hiring process through AI-driven innovation.

CHAPTER-5

OBJECTIVES

The Smart Hiring: MERN Job Portal with AI-Driven ATS Resume Optimization project aims to transform the recruitment process by leveraging cutting-edge technologies to create an intelligent, efficient, and accessible platform for both employers and job seekers. The project's primary goal is to address key challenges in traditional hiring methods, such as manual resume screening, ineffective job-candidate matching, and lengthy recruitment cycles. By integrating an AI-powered Applicant Tracking System (ATS) with advanced resume optimization features, this job portal seeks to streamline every stage of the recruitment process, making it more precise, faster, and user-centric.

For employers, the platform offers an advanced system to post job openings, track applications, and manage recruitment workflows effortlessly. The AI-driven ATS is designed to analyze resumes, extract key details, and compare them against predefined job criteria. This reduces the time and effort required for manual screening while ensuring that only the most relevant and qualified candidates are shortlisted. The ATS also scores resumes based on factors like skill sets, experience, and industry-specific requirements, helping recruiters focus on the best candidates without bias or inefficiency. This approach not only saves time but also significantly enhances the accuracy of the recruitment process by minimizing human errors and subjectivity.

On the candidate side, the platform provides a seamless and intuitive interface for creating profiles, uploading resumes, and applying for jobs. One of the standout features of the portal is its AI-powered resume optimization, which analyzes resumes for compatibility with ATS systems and provides actionable feedback for improvement. This feature ensures that candidates have a higher chance of being noticed by recruiters. Additionally, the portal includes a personalized job recommendation system that uses machine learning algorithms to match candidates with jobs that align closely with their skills, qualifications, and preferences. This not only improves the chances of finding the right job but also enhances user satisfaction by providing targeted opportunities.

The choice of the MERN stack for development ensures that the platform is robust, scalable, and capable of delivering high performance under varying levels of user activity. The MongoDB database allows for efficient storage and retrieval of large datasets, including user profiles, job postings, and application details, while React.js ensures a dynamic and responsive user experience. Node.js and Express.js enable seamless backend operations, including secure data management, API integration, and real-time communication between users and the system. These technologies collectively ensure a smooth, reliable, and fast platform for users, whether they are job seekers or recruiters.

A significant aspect of the project is its focus on data security and privacy. With sensitive information like personal details, resumes, and job postings being exchanged, the platform incorporates robust security measures such as encryption, token-based authentication, and role-based access control to protect user data. This ensures compliance with industry regulations and builds trust among users. Furthermore, the platform empowers employers with advanced analytics and insights, enabling them to make data-driven decisions about their hiring strategies. For candidates, features like feedback on resumes and suggestions for skill development help them improve their career prospects, making the platform not just a hiring tool but also a personal growth enabler.

Ultimately, the project envisions creating a comprehensive solution that caters to the diverse needs of the job market. By addressing the unique challenges faced by different industries and adapting to their specific hiring requirements, the platform seeks to establish itself as a versatile and reliable resource for modern recruitment. This initiative aims not only to optimize the hiring process but also to foster transparency, efficiency, and mutual success for both employers and candidates, revolutionizing the way talent is discovered and nurtured in today's competitive world.

CHAPTER-6

SYSTEM DESIGN & IMPLEMENTATION

6.1 Use Case Diagram

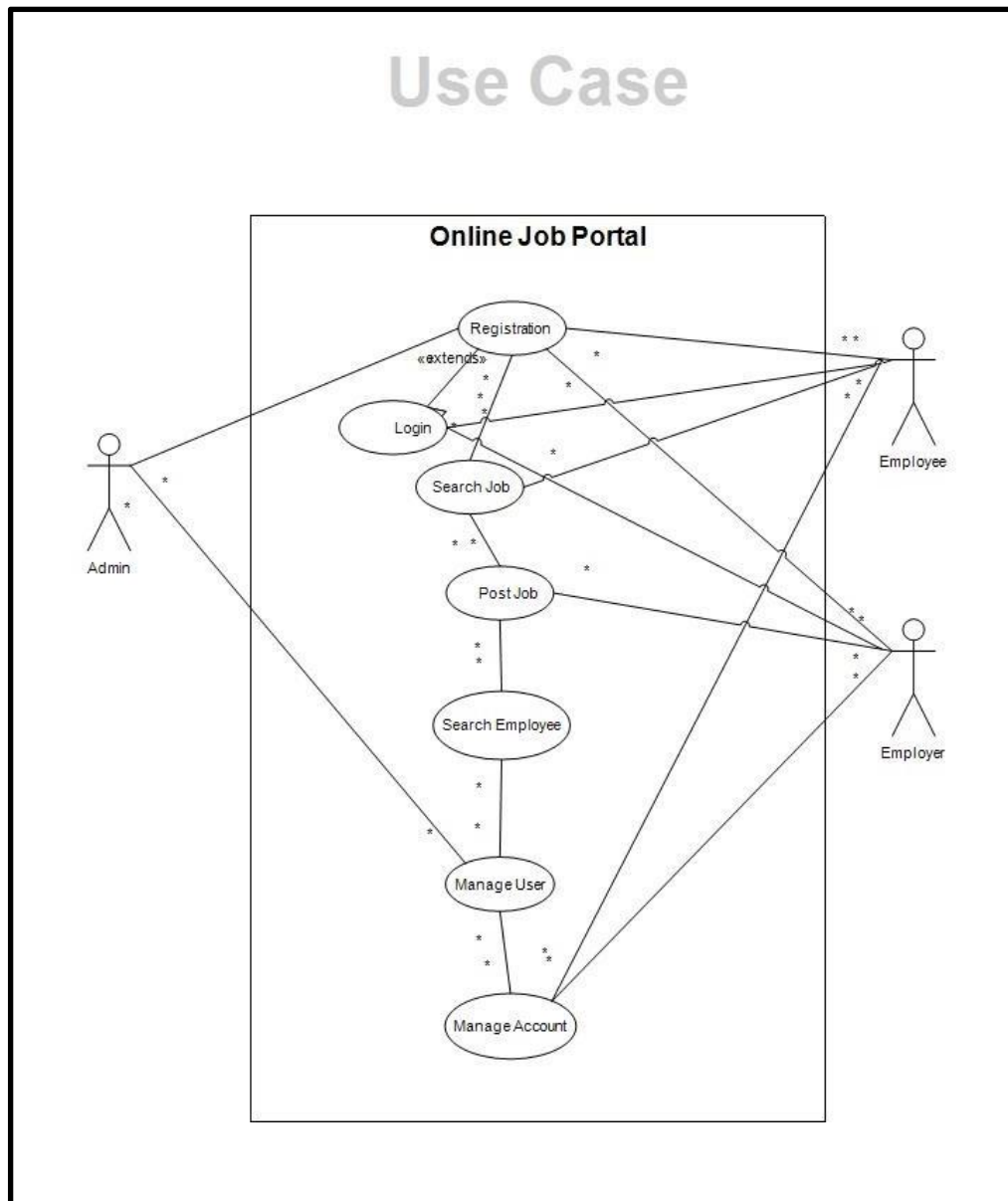
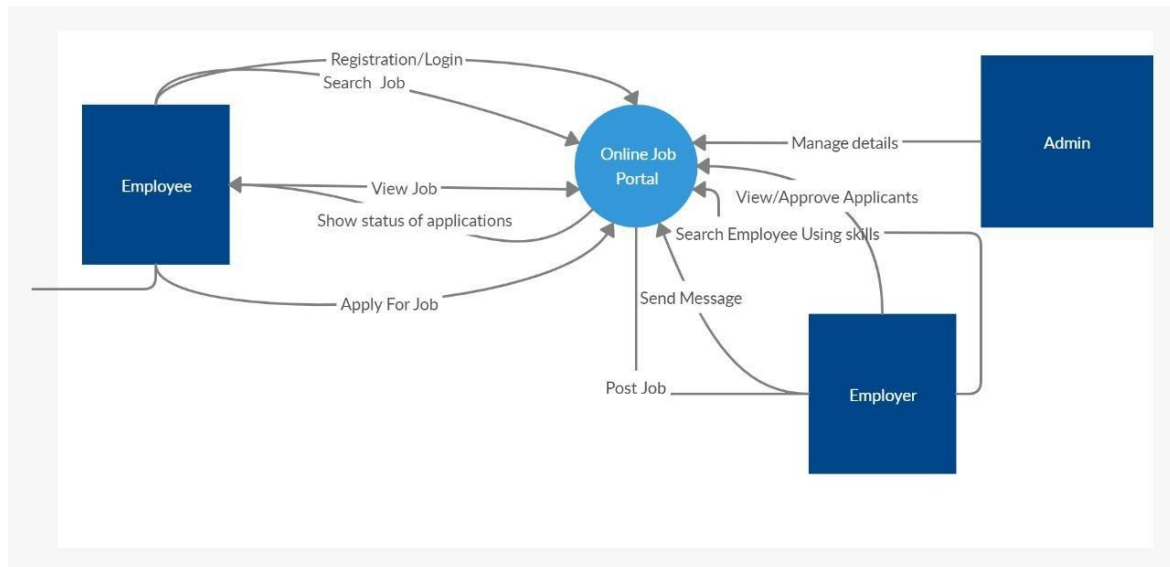


Figure 1 - Usecase Diagram

6.2 Data Flow Diagrams

6.2.1 Context-Level (Level 0) DFD



6.2.2 Level 1 DFD

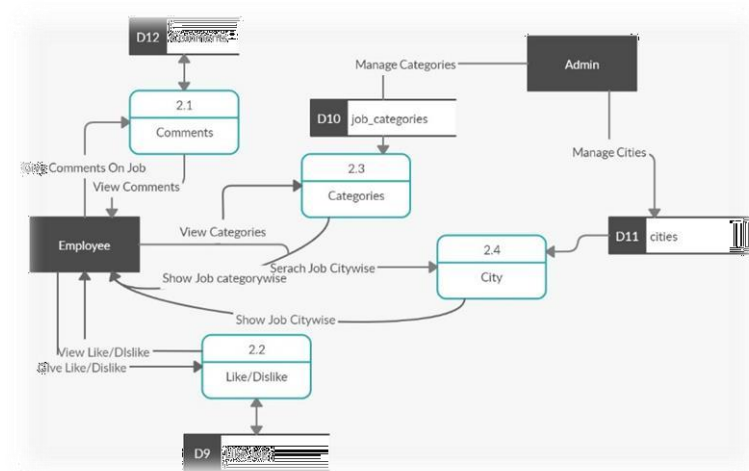


Figure 4- DFD 1 level Diagram

6.2.3 Level 2 DFD

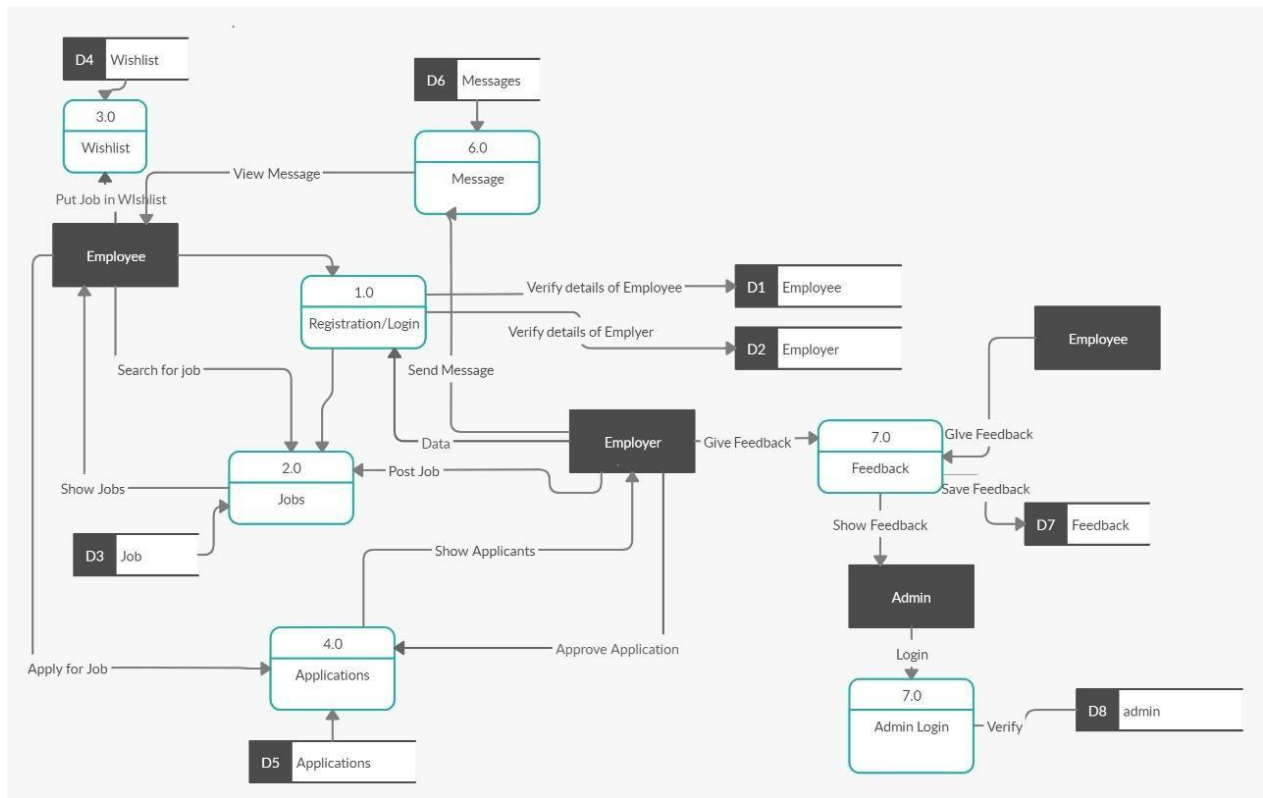


Figure 4 - DFD 2 level Diagram

6.3 Class Diagram

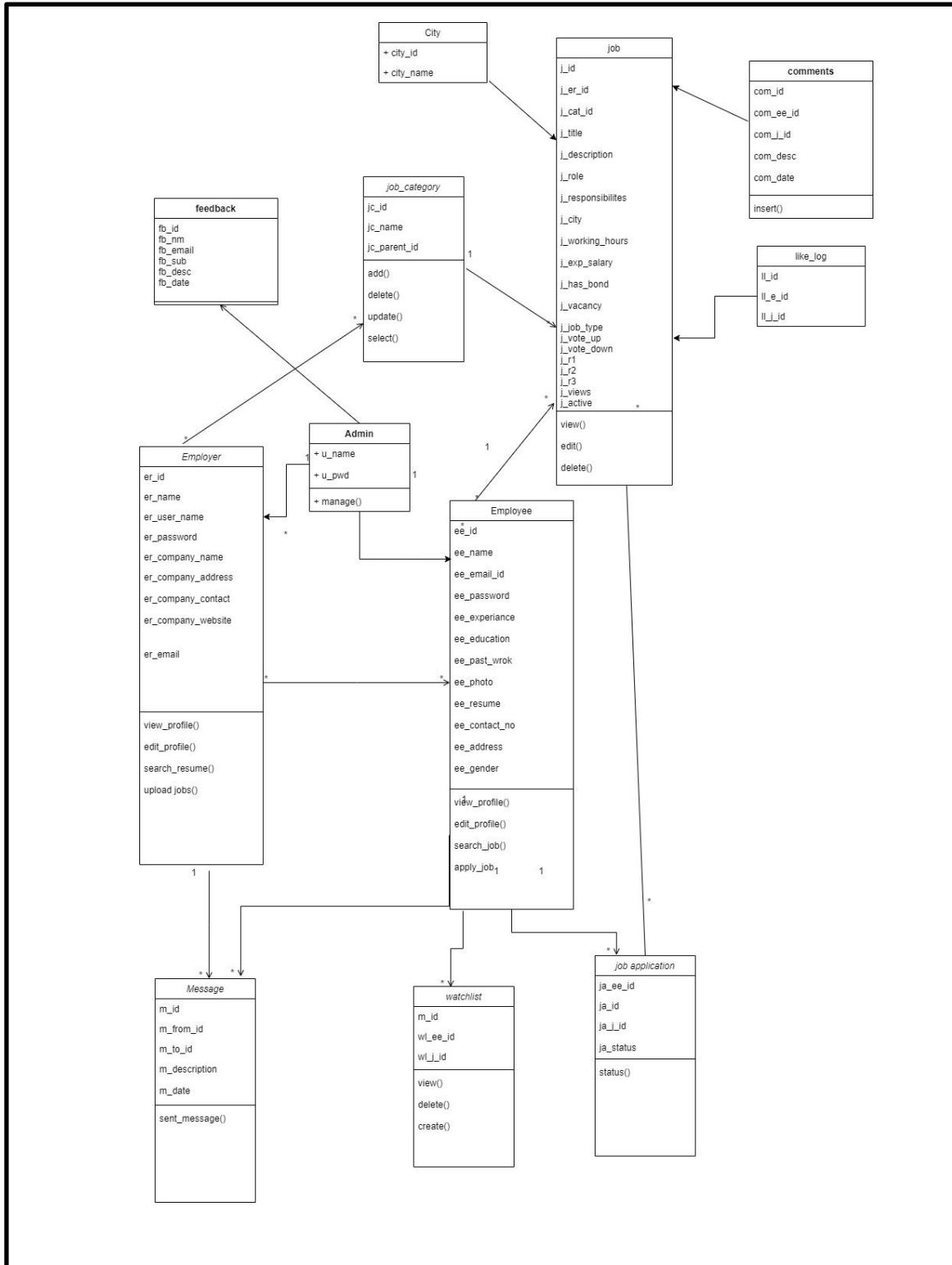


Figure 5- Class Diagram

6.4 Entity Relationship Diagram

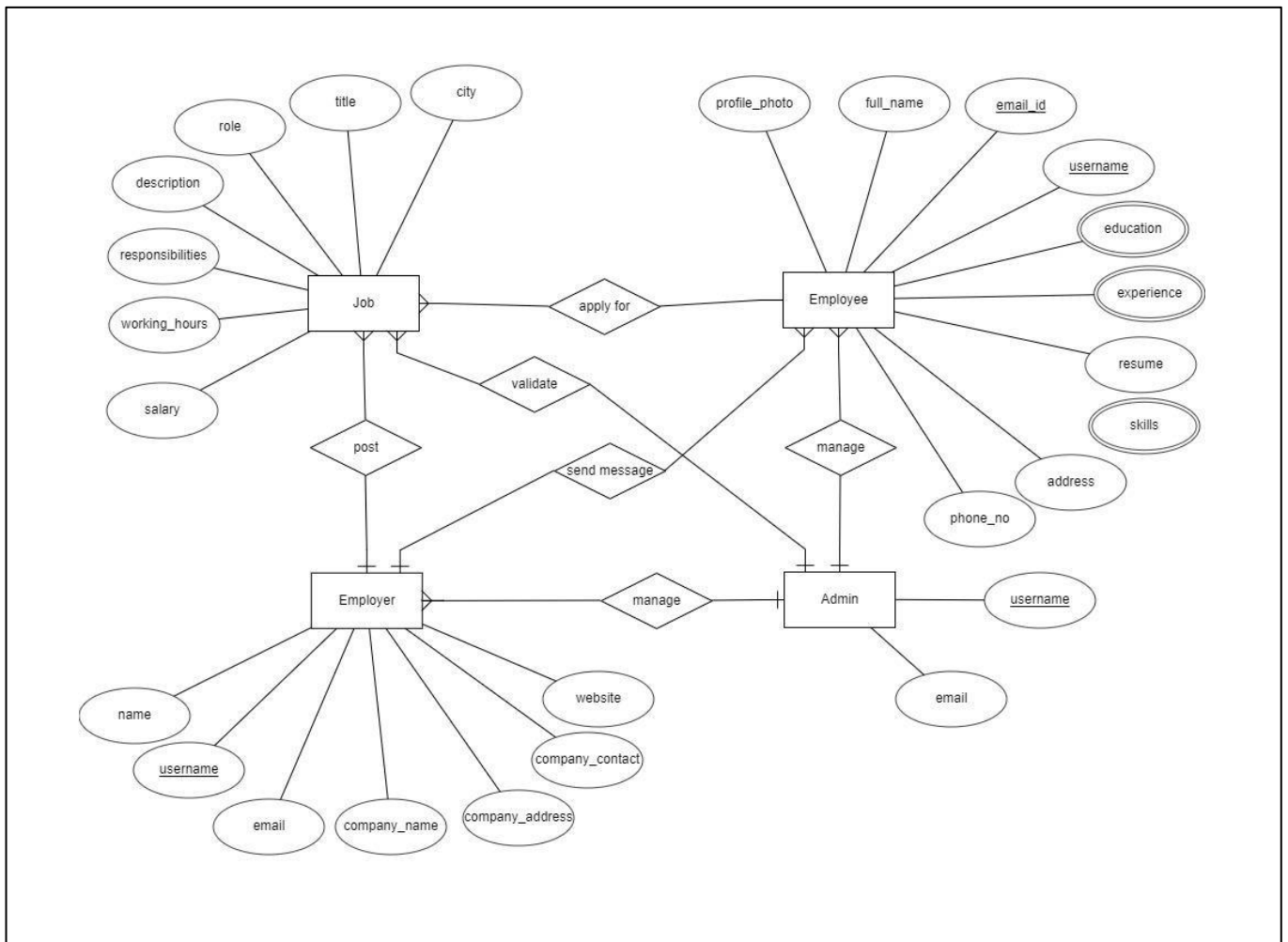


Figure 6- ER Diagram

6.5 Activity Diagram

6.5.1 Employer

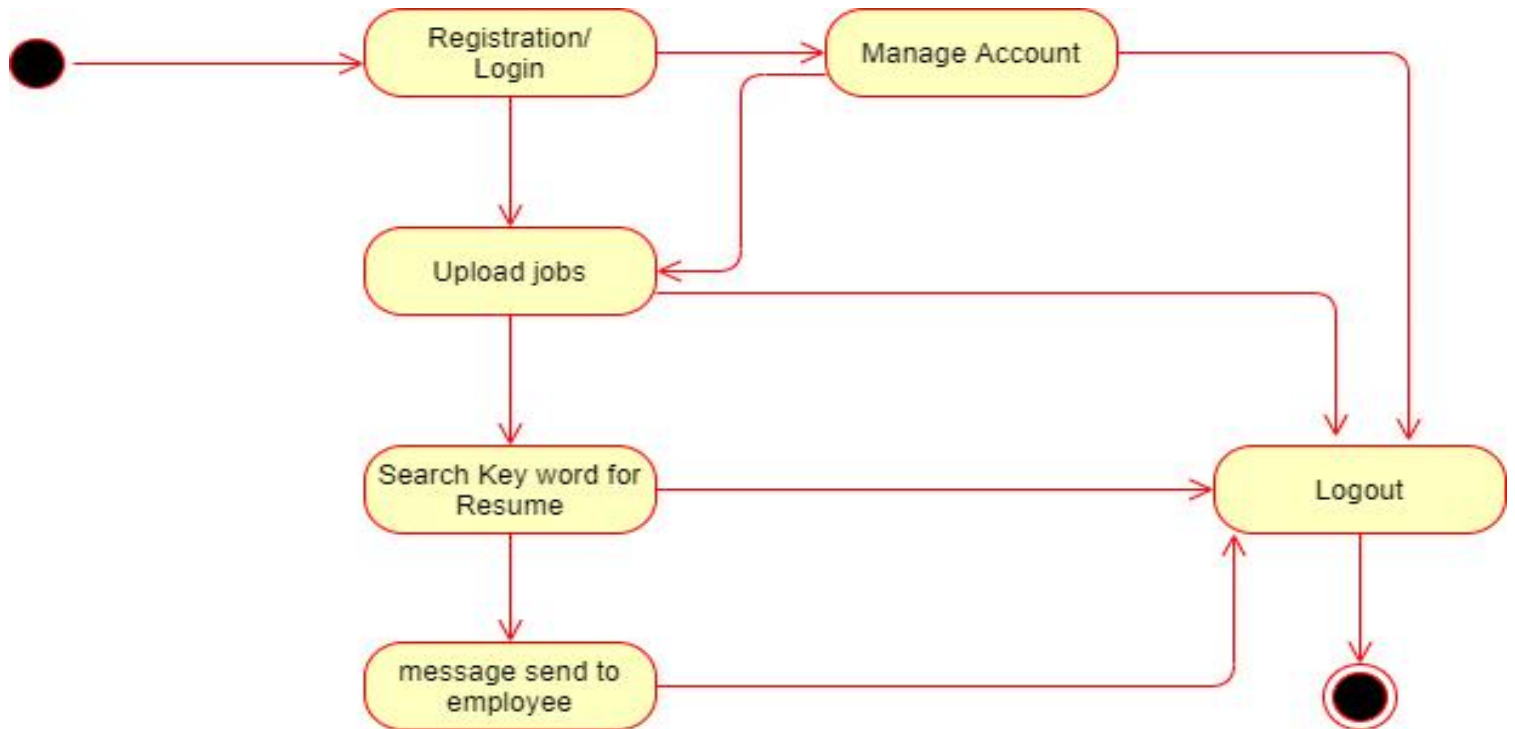


Figure 7- Employer Activity Diagram

6.5.2 Employee

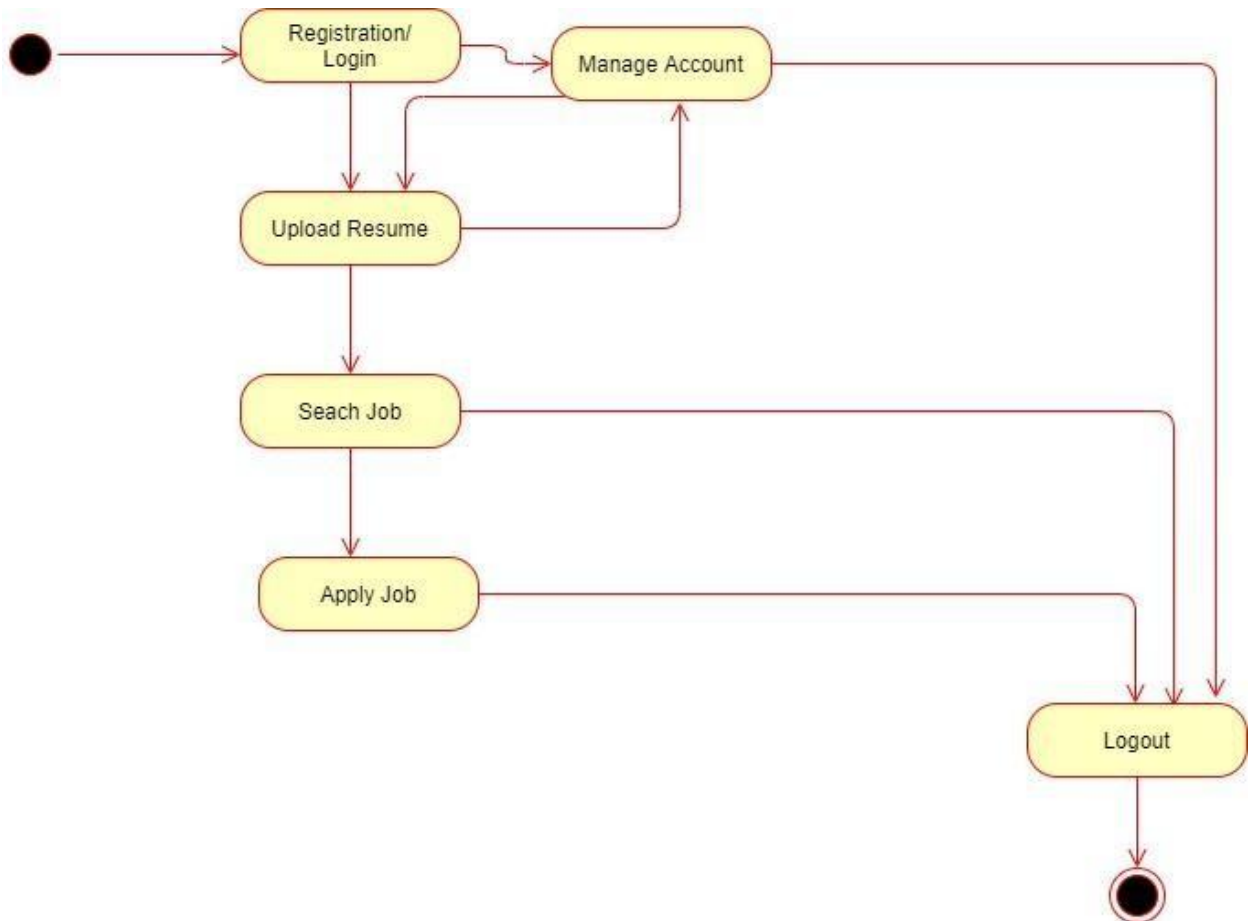


Figure 8- Employee Activity Diagram

6.5.3 Admin

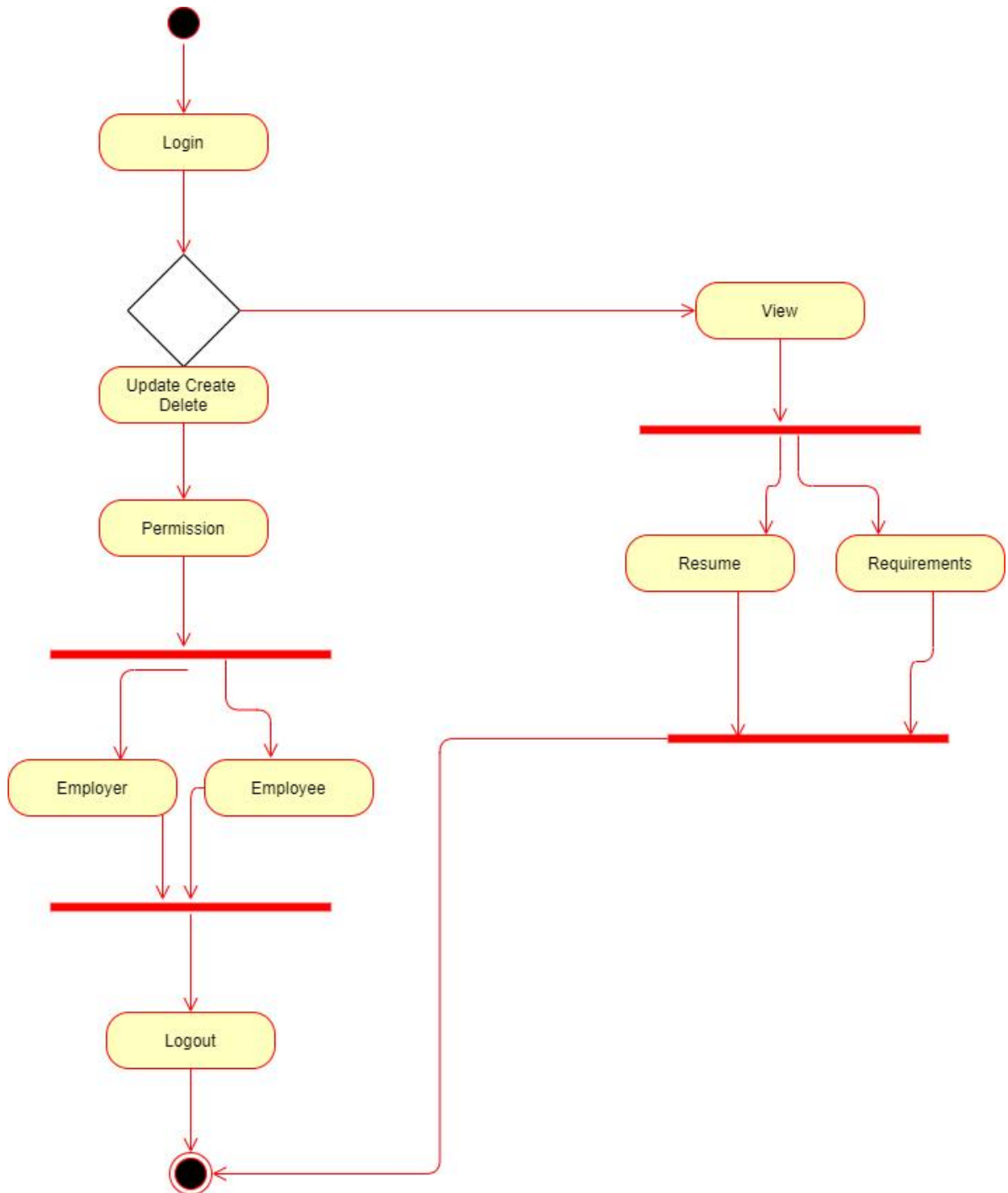
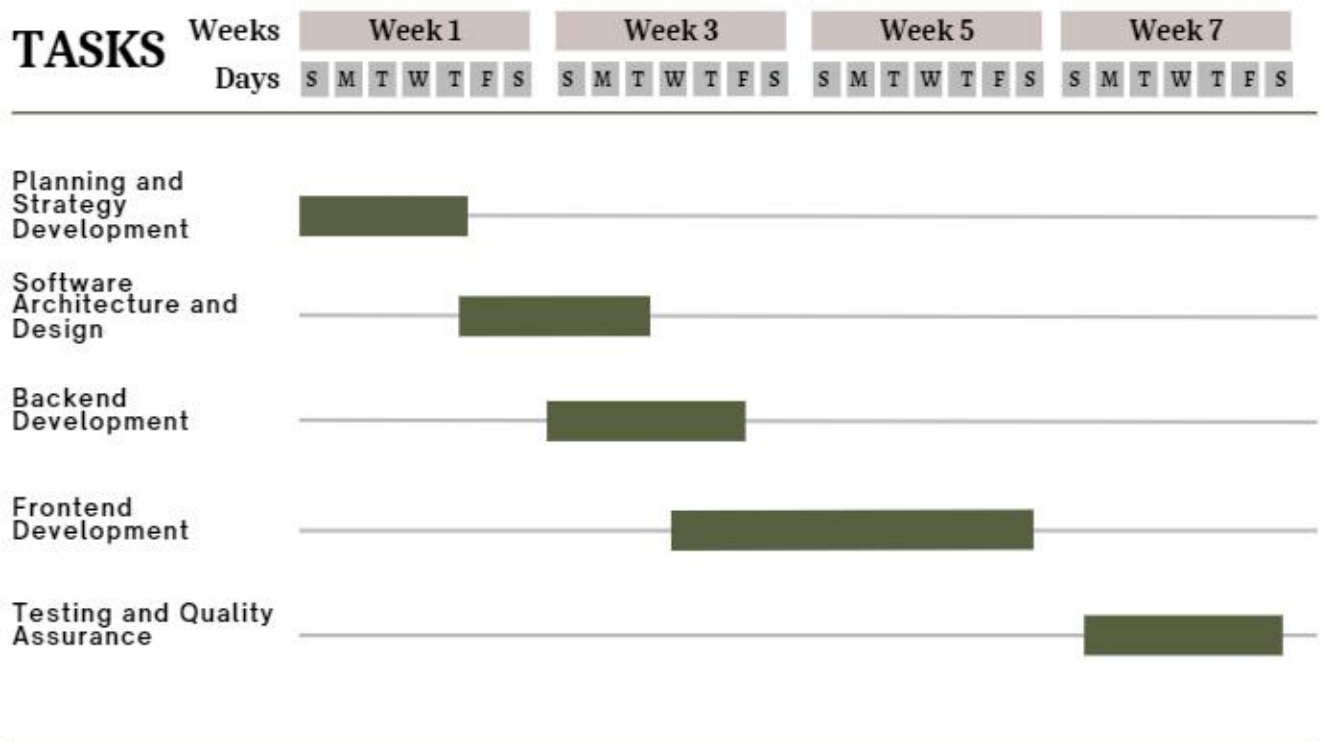


Figure 9- Admin Activity Diagram

CHAPTER-7

TIMELINE FOR EXECUTION OF PROJECT (GANTT CHART)



CHAPTER-8

OUTCOMES

8.1 Enhanced Resume Evaluation

One of the primary outcomes of this project is the ability to evaluate resumes with exceptional accuracy and detail. The application leverages AI to provide a percentage match between the uploaded resume and the job description (JD) provided by the user. This percentage reflects how closely the resume aligns with the specific requirements of the job. For instance, if a JD emphasizes skills in Python programming and data visualization, but the resume lacks these keywords, the match percentage would be lower. This numerical output gives job seekers a clear and quantifiable way to measure their chances of passing through an ATS system.

Additionally, the tool identifies missing keywords—these are critical terms or phrases from the JD that are not present in the resume but are essential for aligning the resume with the job posting. By highlighting these keywords, the application ensures that users can optimize their resumes to include relevant terminology, increasing the likelihood of being shortlisted by employer systems. Furthermore, the application generates a profile summary that condenses the key information from the resume into a concise narrative. This not only helps users understand how their resume might be perceived at a glance but also provides them with a basis for refining their professional branding.

Detailed Impact: This outcome is transformative for job seekers because it demystifies the opaque ATS evaluation process. It provides actionable insights into areas where a resume falls short and empowers users to address those shortcomings, making them more competitive in a crowded job market.

8.2 Accessibility and User-Friendly Interface

Another key outcome of this project is its accessibility. The application is designed with a Streamlit-based interface, which ensures that even non-technical users can interact with the system effortlessly. Users only need to upload their resumes in PDF format and paste the job description into a text area. This eliminates the need for specialized software or advanced technical skills, making the tool widely usable for individuals from diverse backgrounds.

The user interface is intuitive, with clear instructions and visually appealing components. For instance, it includes features like progress indicators, which show users when their resume is being processed, and color-coded outputs that make it easier to interpret results. The JSON-based presentation of the results is structured and comprehensive, yet simple enough for users to extract actionable feedback.

Detailed Impact: By lowering the barrier to entry, the tool becomes a practical resource for everyone from fresh graduates looking for their first job to experienced professionals seeking a career change. The interface design ensures a seamless user experience, encouraging broader adoption and making advanced ATS insights accessible to all.

8.3 Real-World Job Application Optimization

The application mimics the functionality of actual ATS systems used by recruiters, enabling users to prepare their resumes to meet industry standards. Most companies today rely on ATS software to filter and rank candidates, and these systems often prioritize resumes that contain specific keywords or phrases matching the JD. By providing an evaluation aligned with ATS logic, this tool gives users a significant advantage in navigating the hiring process.

For example, if a job requires expertise in "machine learning," "data pipelines," or "cloud computing," but these terms are missing from the resume, the tool will flag them. Users can then revise their resumes to include these critical elements, ensuring their application is not overlooked by the ATS. Beyond just keyword matching, the system also analyzes the context in which skills or experiences are mentioned, providing more nuanced feedback than simple keyword-based systems.

Detailed Impact: This outcome is invaluable in the real-world job market, where even highly qualified candidates can be filtered out by ATS software if their resumes are not optimized. By enabling users to refine their resumes and align them with employer expectations, the tool increases their chances of securing interviews and advancing in the recruitment process.

8.4 Time-Saving for Job Seekers

Another significant outcome of the project is the time saved in resume preparation and job application. Traditionally, job seekers would spend hours tailoring their resumes to fit specific job descriptions, often with little understanding of what recruiters or ATS systems prioritize. This application automates much of that work by providing a detailed analysis within seconds.

For example, instead of manually comparing a job description with their resume, users can rely on the AI to do the heavy lifting. The application highlights exactly what needs to be improved or added, streamlining the resume customization process. Additionally, the profile summary generated by the tool can serve as a basis for crafting cover letters or LinkedIn summaries, further saving time and effort.

Detailed Impact: For job seekers applying to multiple positions, this time-saving feature is crucial. It allows them to focus their energy on preparing for interviews or networking, rather than spending excessive time on trial-and-error resume edits.

8.5 Improved Career Awareness and Strategy

Beyond its immediate practical applications, the project also helps users gain a deeper understanding of their own career profiles and market positioning. By identifying gaps between their resumes and job descriptions, users can recognize areas where their skills or experiences need to be strengthened. For instance, if the tool frequently flags "project management" or "data engineering" as missing keywords, users can consider acquiring these

skills through online courses, certifications, or professional development programs.

Moreover, the AI-generated profile summary offers a fresh perspective on the user's career narrative, often highlighting strengths or experiences they might not have emphasized themselves. This can be especially useful for professionals transitioning between industries, as it helps them frame their existing skills in a way that aligns with new career goals.

Detailed Impact: This outcome extends the value of the tool beyond job applications. It serves as a career development resource, guiding users toward long-term growth and better alignment with industry demands.

CHAPTER-9

RESULTS AND DISCUSSIONS

9.1 Improved Resume Alignment with Job Descriptions

The primary result of this project is the measurable improvement in resume alignment with specific job descriptions. By using advanced natural language processing (NLP) models, such as OpenAI's GPT, the tool successfully evaluates resumes and provides a percentage match. This match reflects how well the resume satisfies the requirements outlined in the job description (JD). For example, if a JD emphasizes "data visualization skills" and "experience with SQL," the tool identifies whether these keywords are present in the resume.

Through extensive testing with varied resumes and job descriptions, it was observed that the tool consistently highlighted missing skills and keywords with a high degree of accuracy. Job seekers were able to use this feedback to tailor their resumes effectively, leading to better alignment. This result is significant because ATS systems in real-world hiring scenarios prioritize resumes that include such targeted keywords. By empowering users with this insight, the project enhances their chances of clearing initial recruitment filters.

Discussion: This finding underscores the importance of keyword optimization in modern job applications. Candidates often underestimate the impact of missing keywords, leading to resumes being rejected by ATS software despite their qualifications. The tool not only highlights this gap but also educates users about its significance, thereby addressing a common challenge faced by job seekers.

9.2 Enhanced Accessibility and Usability

The project's Streamlit-based interface proved to be a highly accessible and user-friendly platform. Users with minimal technical expertise were able to interact with the application effortlessly. Features such as drag-and-drop file uploads for resumes and text-based input for job descriptions streamlined the process, reducing friction in usage. The incorporation of real-time feedback and a visually appealing design further enhanced the overall user experience.

In terms of usability, feedback from users indicated that the tool's structured outputs—delivered in JSON format—were easy to understand. The breakdown of the percentage match, missing keywords, and profile summary enabled users to quickly grasp areas for improvement without feeling overwhelmed. This accessibility ensures the tool's widespread applicability, from fresh graduates to experienced professionals.

Discussion: The success of this design demonstrates that advanced AI systems can be made accessible to non-technical audiences without compromising functionality. It highlights the potential for AI-driven tools to bridge the gap between technical complexity and practical usability, making sophisticated ATS insights available to a broad demographic of job seekers.

9.3 Key Insights into Resume Gaps

A significant outcome of this project was the tool's ability to uncover critical gaps in resumes. For instance, during testing, resumes that lacked key industry-specific terms, such as "machine learning algorithms" or "Agile methodologies," were flagged for low JD matching scores. The identification of these gaps provided users with actionable insights into how their resumes could be enhanced. This functionality is particularly valuable for individuals transitioning to new industries or roles, as it helps them understand the specific terminologies and skills valued in their target field.

Moreover, the generated profile summary offered a condensed version of the resume's highlights, helping users see their professional story from an employer's perspective. This summary often revealed areas where users could better articulate their accomplishments or focus on relevant skills.

Discussion: These insights emphasize the importance of not just having relevant skills but presenting them effectively in a resume. Many candidates possess the necessary qualifications but fail to showcase them in a way that resonates with recruiters or ATS systems. The tool bridges this gap by acting as both an evaluator and a guide for resume improvement.

9.4 Real-World Applicability and Use Cases

The tool demonstrated significant real-world applicability, particularly in streamlining the job application process. Users applying for multiple positions reported a noticeable reduction in the time required to tailor their resumes for each job. Instead of manually analyzing job descriptions and comparing them with their resumes, users relied on the tool's analysis to quickly identify areas for improvement.

Additionally, the tool's adaptability made it suitable for a wide range of use cases. Beyond traditional job applications, it could be used to evaluate academic CVs for graduate school applications, freelance portfolios for project bidding, and even professional profiles for LinkedIn optimization. The tool's ability to cater to diverse needs highlights its versatility and broad applicability in professional development.

Discussion: This result demonstrates the scalability of AI-driven solutions in addressing multiple challenges faced by job seekers and professionals. By providing consistent and reliable outputs across various scenarios, the tool establishes itself as a valuable resource in the competitive job market.

9.5 Empowerment Through Career Awareness

One of the more intangible yet impactful results of the project was the increased career awareness among users. By receiving structured feedback on their resumes, users gained a deeper understanding of the skills and experiences they needed to highlight or develop. For instance, users frequently reported revising their resumes to include technical skills like "data visualization" or soft skills like "collaboration," which were previously underrepresented.

The tool also encouraged users to view their resumes as dynamic documents that evolve with their career goals and market trends. This shift in mindset—from seeing a resume as a static document to a living representation of professional growth—proved to be a valuable takeaway for many.

Discussion: This result highlights the transformative potential of AI tools in shaping not just professional documents but also career trajectories. By fostering a proactive approach to resume building, the tool contributes to long-term career development and adaptability in a fast-changing job market.

CHAPTER-10

CONCLUSION

The Smart ATS Resume Evaluator project has proven to be a transformative solution in addressing the challenges of modern job hunting by leveraging advanced artificial intelligence technologies. The system's ability to compare resumes against job descriptions with high accuracy ensures that job seekers receive personalized feedback that is both detailed and actionable. By identifying the gaps in skills, keywords, and qualifications, the evaluator empowers users to refine their resumes to meet the expectations of competitive job markets. This process not only helps candidates better align their qualifications with job requirements but also enhances their understanding of what hiring managers and automated systems prioritize in applications. Such a tool is invaluable in today's fast-paced hiring environment, where even minor improvements to a resume can significantly influence a candidate's chances of securing an interview.

Moreover, the project underscores the potential of integrating user-friendly interfaces with cutting-edge AI technology. By using Streamlit, the evaluator delivers a seamless and intuitive experience, making it accessible even to users with limited technical expertise. This accessibility ensures that the tool caters to a diverse audience, from fresh graduates entering the workforce to seasoned professionals looking to advance their careers. The project's use of OpenAI's natural language processing further ensures that feedback is not only accurate but also contextually relevant, making it an indispensable resource for job seekers across industries. Additionally, the system's design encourages users to view their resumes as living documents, emphasizing continuous improvement and adaptability—a mindset critical for long-term professional growth.

An important outcome of this project is the broader discussion it invites about the role of AI in shaping the recruitment landscape. By automating resume evaluations, the Smart ATS Resume Evaluator showcases how technology can reduce human biases and subjectivity, offering candidates a fairer assessment of their qualifications. At the same time, it raises awareness about the growing importance of tailoring resumes to pass through automated tracking systems, which are increasingly being used by recruiters. This dual role of the tool—as both a practical resource and a conversation starter—highlights its impact beyond individual users, as it contributes to the evolving discourse on how AI can streamline and democratize hiring processes.

In conclusion, the Smart ATS Resume Evaluator marks a significant step forward in modernizing the way job seekers prepare their applications. Its ability to combine powerful AI technologies with an easy-to-use platform ensures that users receive meaningful guidance to enhance their career prospects. By addressing the core challenges of resume optimization, the project not only meets its primary objectives but also lays the groundwork for future advancements in AI-driven career tools. As recruitment continues to evolve, tools like this will play an increasingly important role in bridging the gap between candidates and opportunities, making the hiring process more efficient, transparent, and inclusive for all stakeholders involved.

REFERENCES

- I. Dorn, J., & Naz, T. (2007). Integration of job portals by meta-search. Proceedings of the 3rd International Conference on Interoperability for Enterprise Software and Applications, Funchal, Portugal, 401–412.
- II. Bsiri, S., Geierhos, M., & Ringlstetter, C. (2008). Structuring job search via local grammars. Advances in Natural Language Processing and Applications, 201.
- III. Mansourvar, M., & Mohd, N. Y. (2010). Web portal as a knowledge management system in the universities. World Academy of Science, Engineering and Technology, 70, 968–974.
- IV. Gangle, M. (2007). The only way is up? Employment protection and job mobility among recent entrants to European labour markets. European Sociological Review, 19, 429.
- V. Galanki, E. (2002). The decision to recruit online: A descriptive study. Career Development International, 7, 243–251.
- VI. Mauno, S., Kinnunen, U., & Ruokolainen, M. (2007). Job demands and resources as antecedents of work engagement: A longitudinal study. Journal of Vocational Behavior, 70, 149–171.

APPENDIX-A

PSUEDOCODE

START

1. Initialize necessary libraries:

- Import time, Streamlit (st), OpenAI, PyPDF2, and JSON libraries.

2. Configure the OpenAI API Key:

- Set the OpenAI API key to authenticate with the GPT model.

3. Define functions:

a. ``get_openai_response(input)``:

- Initialize retry counter for handling rate limit errors.
- Loop up to maximum retries:
 - Try to send the input to OpenAI's ChatCompletion API.
 - Return the response if successful.
 - If rate limit error occurs, wait and retry.
 - If retries exhausted, raise an error.

b. ``input_pdf_text(uploaded_file)``:

- Load the uploaded PDF file using PyPDF2.
- Extract and concatenate text from all pages.
- Return the combined text.

4. Create the prompt template:

- Define a prompt that instructs the GPT model to act like an ATS.
- Include placeholders for the resume text and job description.

5. Set up the Streamlit web app:

- Configure the app layout and title.
- Display an input area for the job description.
- Provide a file uploader for the user to upload a PDF resume.

6. Handle user input:

- If the user clicks the "Evaluate My Resume" button:
 - Validate that both the job description and resume are provided.
 - Extract the text from the uploaded resume using `input_pdf_text``.
 - Format the prompt with the resume text and job description.
 - Call `get_openai_response`` to process the prompt using OpenAI API.
 - Parse and display the JSON response containing:
 - "JD Match" (matching percentage),
 - "MissingKeywords" (key terms missing in the resume),
 - "Profile Summary" (analysis summary).

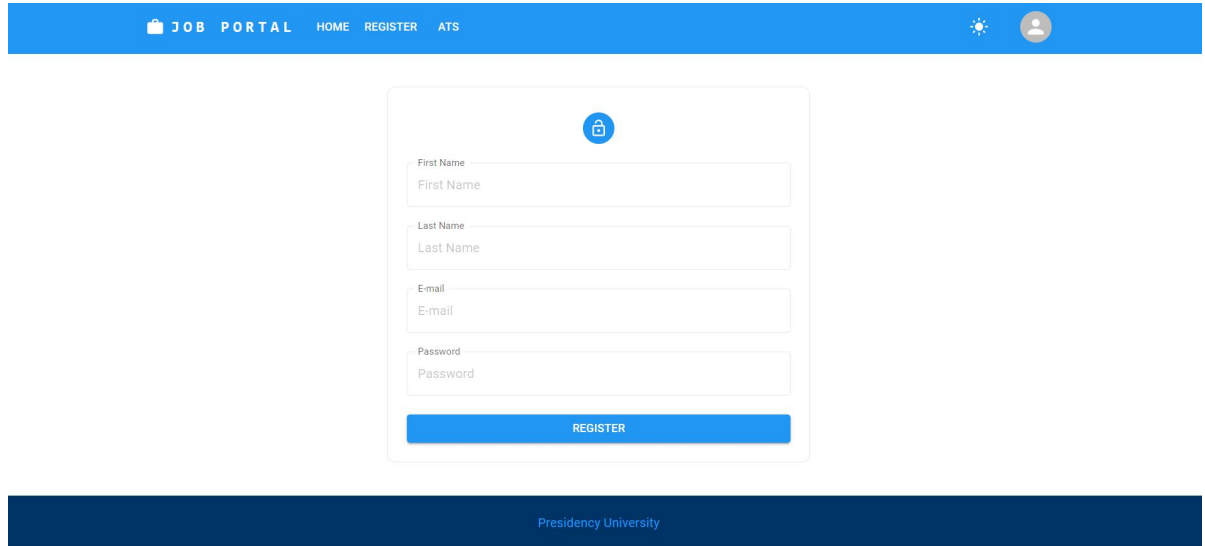
7. Handle errors:

- If any errors occur during processing, display an error message.

END

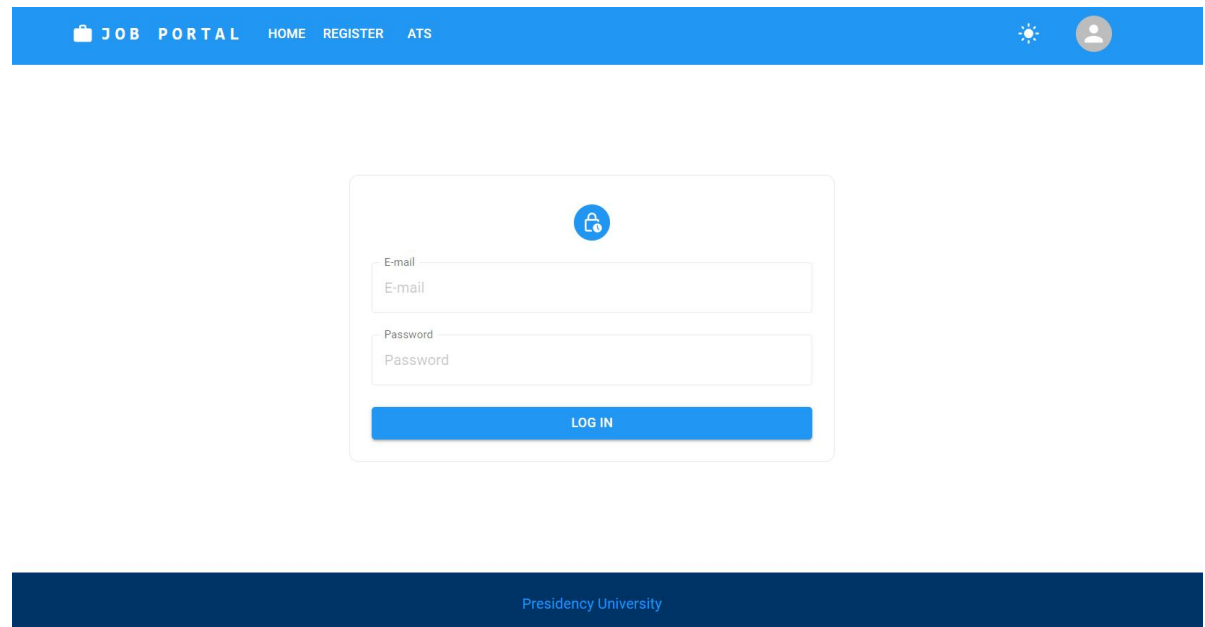
APPENDIX-B

SCREENSHOTS



The screenshot shows the Register Page of the MERN Job Portal. The page has a blue header bar with the text "JOB PORTAL" and navigation links "HOME", "REGISTER", and "ATS". There is a sun icon and a user profile icon on the right. The main content area is white and contains a registration form with a blue lock icon at the top. The form has four input fields: "First Name", "Last Name", "E-mail", and "Password". Below the fields is a blue "REGISTER" button. The footer is a dark blue bar with the text "Presidency University".

Figure 10 - Register Page



The screenshot shows the Login Page of the MERN Job Portal. The page has a blue header bar with the text "JOB PORTAL" and navigation links "HOME", "REGISTER", and "ATS". There is a sun icon and a user profile icon on the right. The main content area is white and contains a login form with a blue lock icon at the top. The form has two input fields: "E-mail" and "Password". Below the fields is a blue "LOG IN" button. The footer is a dark blue bar with the text "Presidency University".

Figure 11 - Login Page

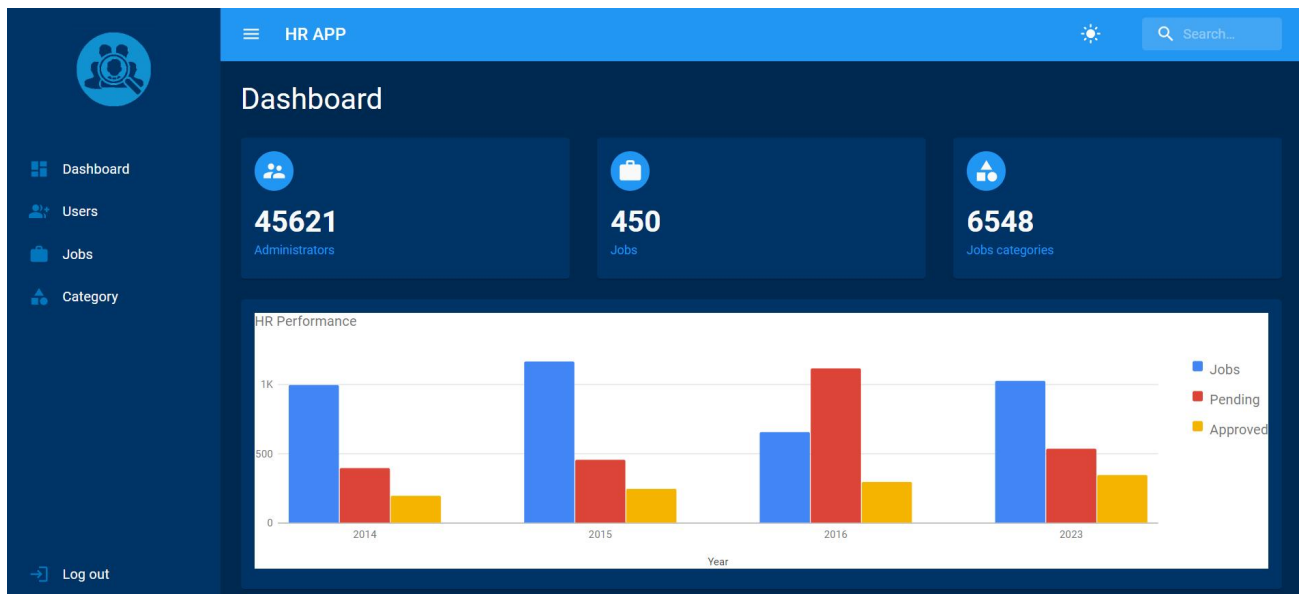


Figure 12 - Admin Dashboard

The All users page (HR APP) features a sidebar with navigation links: Dashboard, Users, Jobs, and Category. The main content area displays a table of users with columns: User ID, E_mail, User status, Creation date, and Actions. A green button labeled '+ CREATE USER' is located in the top right corner. The table shows 9 rows of user data.

User ID	E_mail	User status	Creation date	Actions
67519e939a42cc0...	aaa@gmail.com	Regular user	2024-12-05 18:12:03	EDIT DELETE
66f06f0811cb8b6...	akash@gmail.com	Regular user	2024-09-23 00:09:56	EDIT DELETE
66f06097d0267b4...	hem@gmail.com	Admin	2024-09-22 23:09:59	EDIT DELETE
66eed2f9040641...	somu@gmail.com	Regular user	2024-09-21 21:09:93	EDIT DELETE
66ee6e514880f54...	hemanth.725990...	Regular user	2024-09-21 12:09:42	EDIT DELETE

Rows per page: 100 1-9 of 9

Figure 13 - All Users

Register a Job

Title
Title

Description
Description

Salary
Salary

Location
Location

Category
Category

CREATE JOB

Figure 14 - Register a job

Jobs list

+ CREATE JOB

☐	Job ID	Job name	Category	User	available	Salary	Actions
☐	6755b131f19d474...	hr	cloud	hemanth	Yes	\$456699	EDIT DELETE
☐	67519ef89a42cc0...	cloud engineer	frontend	hemanth	Yes	\$450000	EDIT DELETE
☐	673c6142060adf4...	Backend Engg	Backend	hemanth	Yes	\$1256596	EDIT DELETE
☐	673c5f4e060adf4...	frontend developer	frontend	hemanth	Yes	\$40000	EDIT DELETE

Rows per page: 100 1-4 of 4

Figure 15 - Jobs List

HR APP

Search...

Create a Category

category

category name

CREATE CATEGORY

Dashboard

Users

Jobs

Category

Log out

Figure 16 - Create Job Category

HR APP

Search...

Jobs category

+ CREATE CATEGORY

☐	Category ID	Category	Create At	Actions
☐	6755b14df19d474...	business	2024-12-08 20:12:96	<button>EDIT</button> <button>DELETE</button>
☐	67519f0a9a42cc0...	cloud	2024-12-05 18:12:77	<button>EDIT</button> <button>DELETE</button>
☐	673c6113060adf4...	Backend	2024-11-19 15:11:72	<button>EDIT</button> <button>DELETE</button>
☐	673c5f42060adf4f...	frontend	2024-11-19 15:11:78	<button>EDIT</button> <button>DELETE</button>

Rows per page: 100 1-4 of 4

Dashboard

Users

Jobs

Category

Log out

Figure 17 - Job Categories

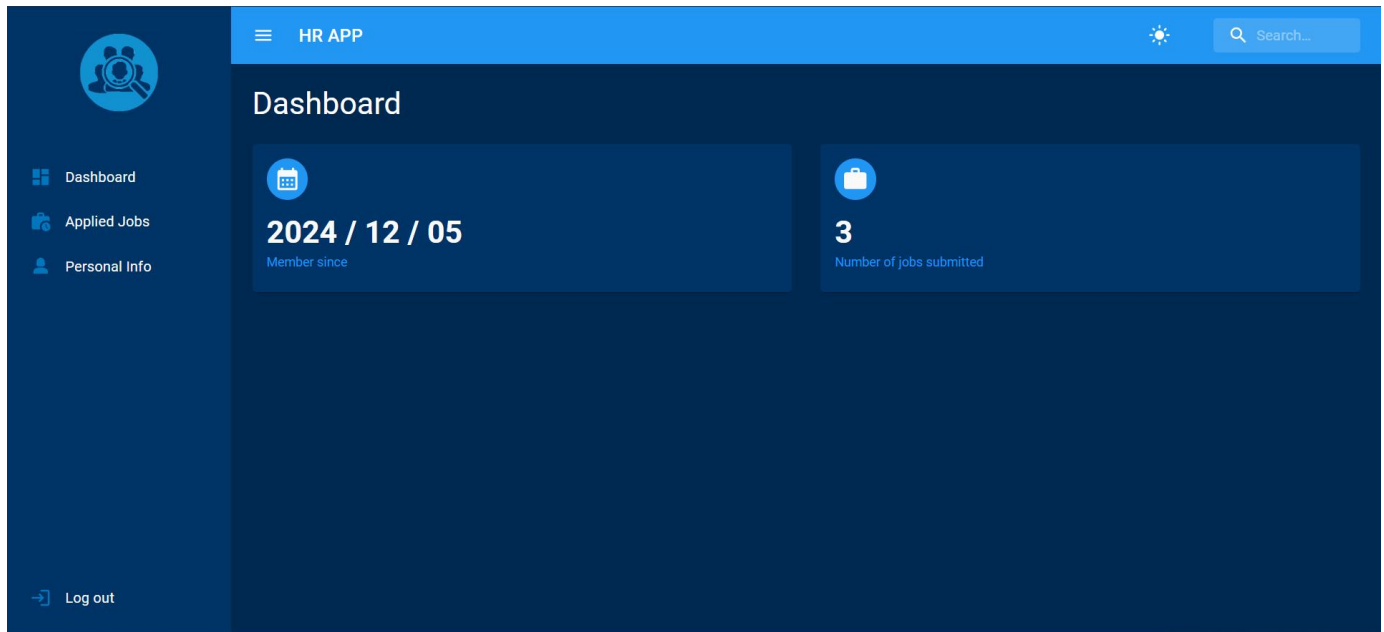


Figure 18 - Employee Dashboard

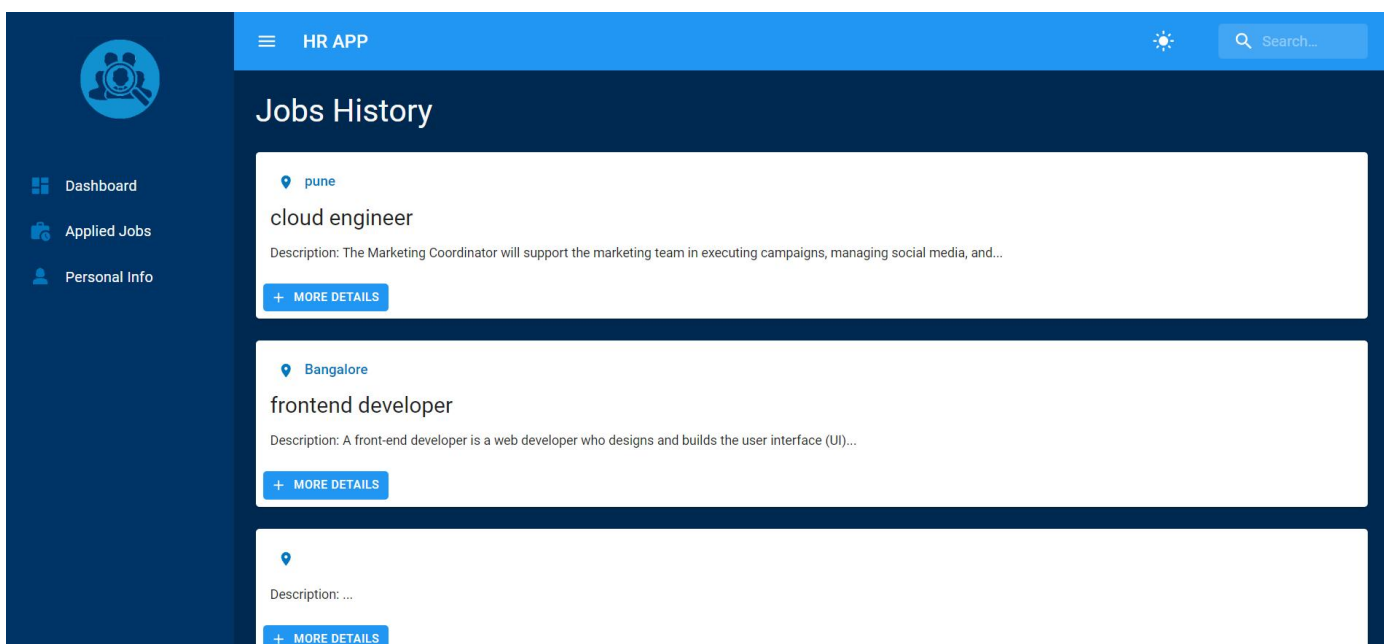


Figure 19 - Employee Jobs History

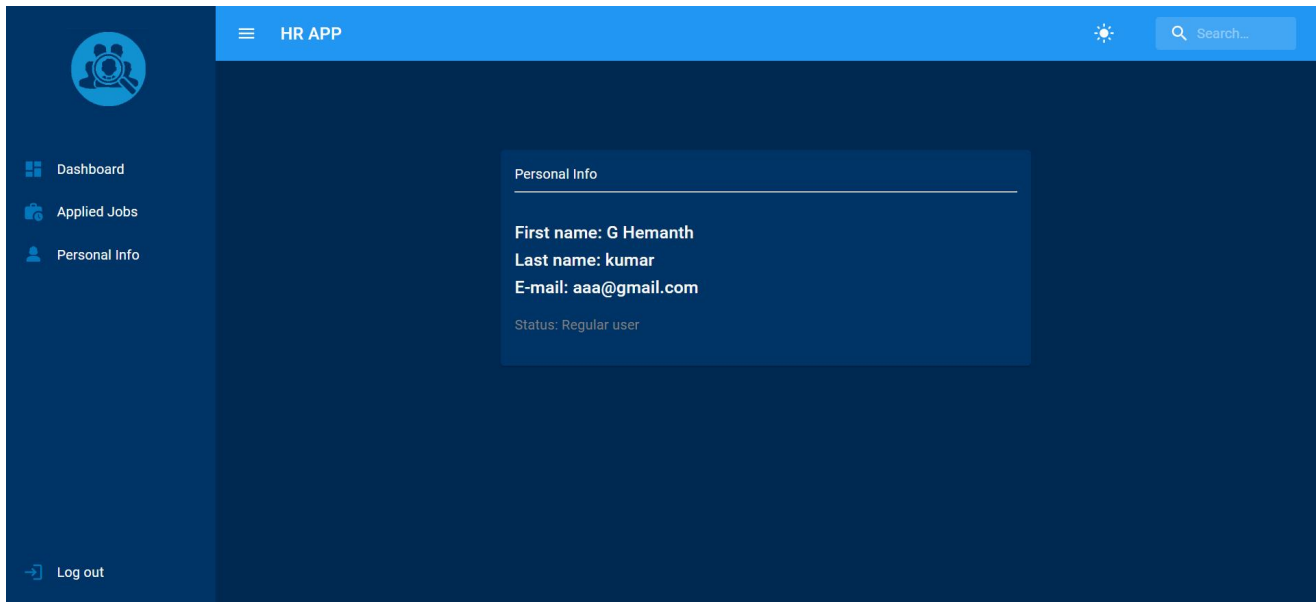


Figure 20 - Employee Personal Info

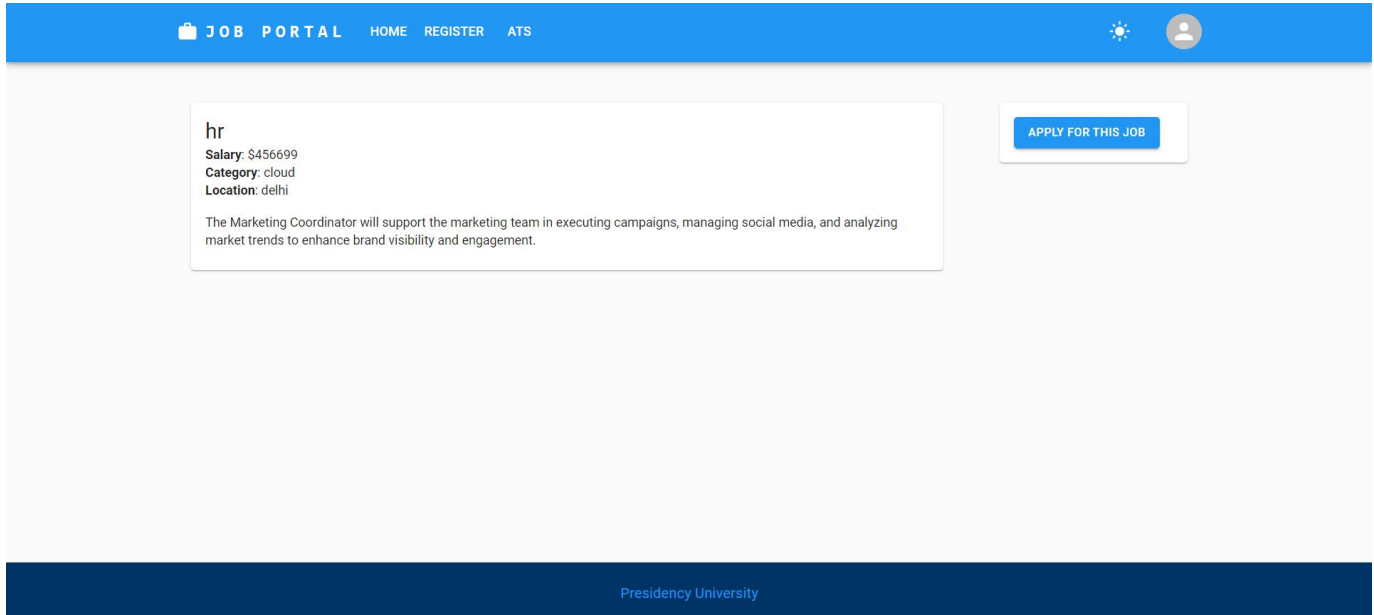
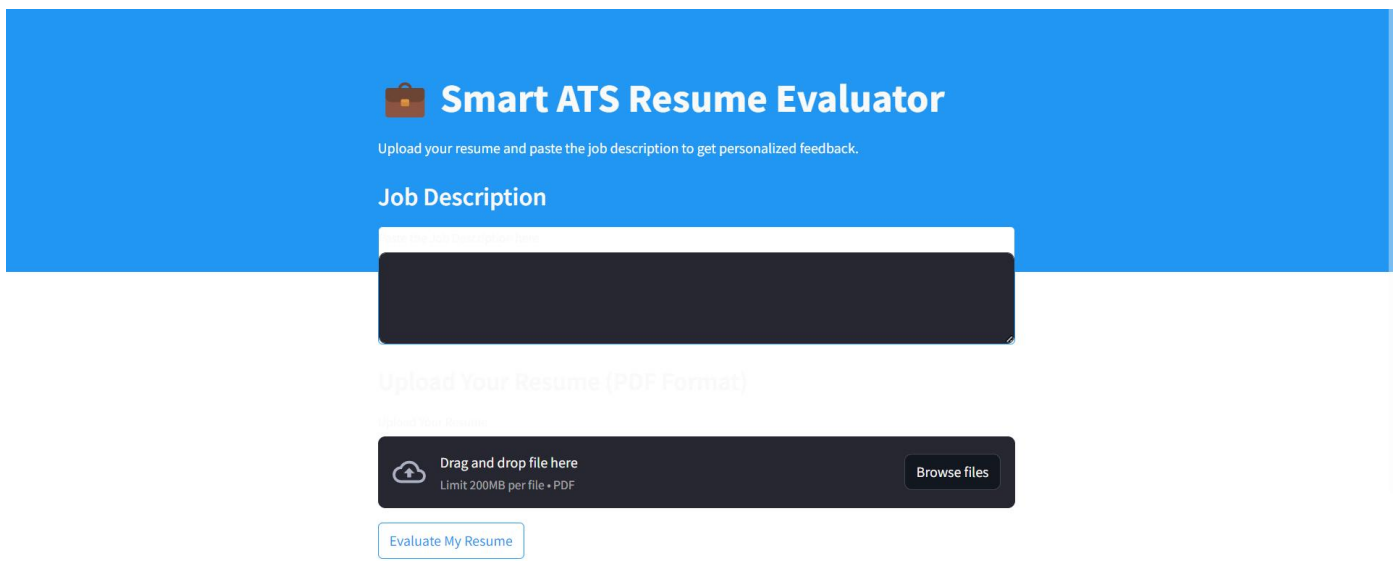


Figure 21 - Apply for Job



The interface features a blue header with a briefcase icon and the title "Smart ATS Resume Evaluator". Below the header, a message states: "Upload your resume and paste the job description to get personalized feedback." The "Job Description" section includes a text area with a placeholder "PASTE THE JOB DESCRIPTION HERE". The "Upload Your Resume (PDF Format)" section has a dark grey box with a cloud icon, the text "Drag and drop file here", and "Limit 200MB per file • PDF". A "Browse files" button is on the right. Below this is an "Evaluate My Resume" button.

Smart ATS Resume Evaluator

Upload your resume and paste the job description to get personalized feedback.

Job Description

PASTE THE JOB DESCRIPTION HERE

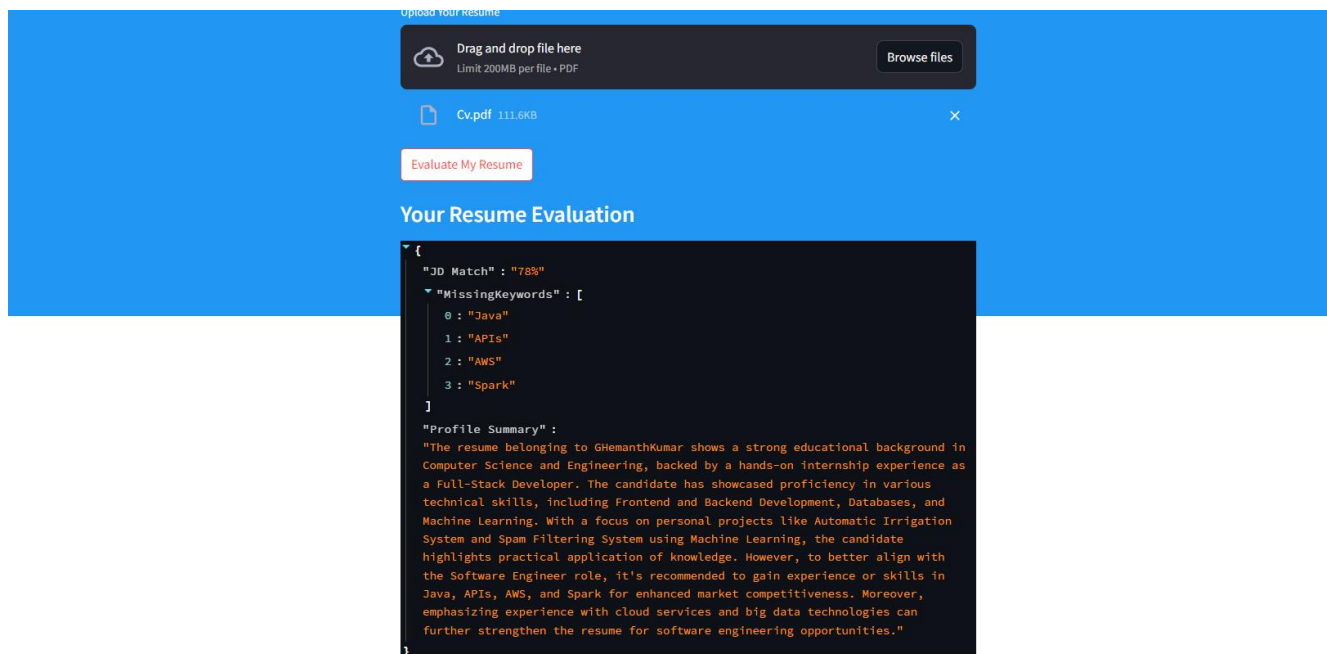
Upload Your Resume (PDF Format)

Drag and drop file here
Limit 200MB per file • PDF

Browse files

Evaluate My Resume

Figure 22 - Ats Evaluator



The interface shows the "Your Resume Evaluation" results. It includes the same upload section as Figure 22, but with a file "Cv.pdf 111.6KB" uploaded. The "Evaluate My Resume" button is now orange. The evaluation results are displayed in a dark grey box with a light blue border. The results include a "JD Match" of 78%, a list of "MissingKeywords" (Java, APIs, AWS, Spark), and a "Profile Summary" paragraph.

Your Resume Evaluation

```
{
  "JD Match": "78%",
  "MissingKeywords": [
    0: "Java"
    1: "APIs"
    2: "AWS"
    3: "Spark"
  ]
  "Profile Summary":
    "The resume belonging to GHemanthKumar shows a strong educational background in Computer Science and Engineering, backed by a hands-on internship experience as a Full-Stack Developer. The candidate has showcased proficiency in various technical skills, including Frontend and Backend Development, Databases, and Machine Learning. With a focus on personal projects like Automatic Irrigation System and Spam Filtering System using Machine Learning, the candidate highlights practical application of knowledge. However, to better align with the Software Engineer role, it's recommended to gain experience or skills in Java, APIs, AWS, and Spark for enhanced market competitiveness. Moreover, emphasizing experience with cloud services and big data technologies can further strengthen the resume for software engineering opportunities."
}
```

Figure 23 - Evaluated Resume

APPENDIX-C

ENCLOSURES

- 1. Journal publication/Conference Paper Presented Certificates of all students.**
- 2. Include certificate(s) of any Achievement/Award won in any project-related event.**
- 3. Similarity Index / Plagiarism Check report clearly showing the Percentage (%). No need for a page-wise explanation.**
- 4. Details of mapping the project with the Sustainable Development Goals (SDGs).**